

torch.stft#

<https://pytorch.org/docs/stable/generated/torch.stft.html>

Description:

Short-time Fourier transform (STFT). From version 1.8.0, `return_complex` must always be given explicitly for real inputs and `return_complex=False` has been deprecated. Strongly prefer `return_complex=True` as in a future pytorch release, this function will only return complex tensors. Note that `torch.view_as_real()` can be used to recover a real tensor with an extra last dimension for real and imaginary components.

Function Signature

```
torch.stft(input, n_fft, hop_length=None, win_length=None,  
window=None, center=True, pad_mode='reflect',  
normalized=False, onesided=None, return_complex=None,  
align_to_window=None)[source]#
```

Parameters

Returns

A tensor containing the STFT result with shape (B?, N, T, C?) where

Return type

Returns:

Tensor

Notes & Warnings

WARNING

Warning From version 1.8.0, `return_complex` must always be given explicitly for real inputs and `return_complex=False` has been deprecated. Strongly prefer `return_complex=True` as in a future pytorch release, this function will only return complex tensors. Note that `torch.view_as_real()` can be used to recover a real tensor with an extra last dimension for real and imaginary components.

WARNING

Warning From version 2.1, a warning will be provided if a window is not specified. In a future release, this attribute will be required. Not providing a window currently defaults to using a rectangular window, which may result in undesirable artifacts. Consider using tapered windows, such as `torch.hann_window()`.

WARNING

Warning This function changed signature at version 0.4.1. Calling with the previous signature may cause error or return incorrect result.

Detailed Documentation

Documentation

Short-time Fourier transform (STFT).

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Note that `torch.view_as_real()` can be used to recover a real tensor with an extra last dimension for real and imaginary components.

From version 2.1, a warning will be provided if a window is not specified. In a future release, this attribute will be required. Not providing a window currently defaults to using a rectangular window, which may result in undesirable artifacts. Consider using tapered windows, such as `torch.hann_window()`.

The STFT computes the Fourier transform of short overlapping windows of the input. This giving frequency components of the signal as they change over time. The interface of this function is modeled after (but not a drop-in replacement for) librosa stft function.

Ignoring the optional batch dimension, this method computes the following expression:

where m is the index of the sliding window, and ω is the frequency

$$0 \leq \omega < n_{\text{fft}} \quad \vee \quad 0 \leq \omega < \lfloor n_{\text{fft}} / 2 \rfloor + 10 \quad \text{for } \text{onesided}=\text{False}, \text{ or}$$
$$0 \leq \omega < \lfloor n_{\text{fft}} / 2 \rfloor + 10 \quad \vee \quad 0 \leq \omega < \lfloor n_{\text{fft}} / 2 \rfloor + 1 \quad \text{for } \text{onesided}=\text{True}.$$

input must be either a 1-D time sequence or a 2-D batch of time sequences.

If `hop_length` is `None` (default), it is treated as equal to $\text{floor}(n_{\text{fft}} / 4)$.

If `win_length` is `None` (default), it is treated as equal to `n_fft`.

window can be a 1-D tensor of size `win_length`, e.g., from `torch.hann_window()`. If window is `None` (default), it is treated as if having 1 everywhere in the window. If $\text{win_length} < n_{\text{fft}}$, window will be

padded on both sides to length n_{fft} before being applied.

If `center` is `True` (default), input will be padded on both sides so that the t -th frame is centered at time $t \times \text{hop_length}$. Otherwise, the t -th frame begins at time $t \times \text{hop_length}$.

`pad_mode` determines the padding method used on input when `center` is `True`. See `torch.nn.functional.pad()` for all available options. Default is "reflect".

If `onesided` is `True` (default for real input), only values for ω in $[0, 1, 2, \dots, \lfloor \frac{n_{\text{fft}}}{2} \rfloor + 1]$ are returned because the real-to-complex Fourier transform satisfies the conjugate symmetry, i.e., $X[m, \omega] = X[m, n_{\text{fft}} - \omega]^*$. Note if the input or window tensors are complex, then onesided output is not possible.

If `normalized` is `True` (default is `False`), the function returns the normalized STFT results, i.e., multiplied by $(\text{frame_length})^{-0.5}$.

If `return_complex` is `True` (default if input is complex), the return is a $\text{input.dim}() + 1$ dimensional complex tensor. If `False`, the output is a $\text{input.dim}() + 2$ dimensional real tensor where the last dimension represents the real and imaginary components.

Returns either a complex tensor of size $(\ast \times N \times T) \times N$ if `return_complex` is true, or a real tensor of size $(\ast \times N \times T \times 2) \times N$ if `return_complex` is false. Where $\ast$ is the optional batch size of input, N is the number of frequencies where STFT is applied and T is the total number of frames used.

This function changed signature at version 0.4.1. Calling with the previous signature may cause error or return incorrect result.

`input` (Tensor) – the input tensor of shape (B, L) where B is an optional batch

dimension

n_fft (int) – size of Fourier transform

hop_length (int, optional) – the distance between neighboring sliding window frames.
Default: None (treated as equal to $\text{floor}(n_fft / 4)$)

win_length (int, optional) – the size of window frame and STFT filter. Default: None
(treated as equal to n_fft)

window (Tensor, optional) – the optional window function. Shape must be 1d and $\leq n_fft$
Default: None (treated as window of all 1's)

center (bool, optional) – whether to pad input on both sides so that the t -th frame is
centered at time $t \times \text{hop_length}$. Default: True

pad_mode (str, optional) – controls the padding method used when center is True.
Default: "reflect"

normalized (bool, optional) – controls whether to return the normalized STFT results
Default: False

onesided (bool, optional) – controls whether to return half of results to avoid
redundancy for real inputs. Default: True for real input and window, False otherwise.

return_complex (bool, optional) –

whether to return a complex tensor, or a real tensor with an extra last dimension for the
real and imaginary components.

Changed in version 2.0: return_complex is now a required argument for real inputs, as
the default is being transitioned to True.

Deprecated since version 2.0: `return_complex=False` is deprecated, instead use `return_complex=True` Note that calling `torch.view_as_real()` on the output will recover the deprecated output format.

`B?` is an optional batch dimension from the input.

`N` is the number of frequency samples, $(n_fft // 2) + 1$ for `onesided=True`, or otherwise `n_fft`.

`T` is the number of frames, $1 + L // \text{hop_length}$ for `center=True`, or $1 + (L - n_fft) // \text{hop_length}$ otherwise.

`C?` is an optional length-2 dimension of real and imaginary components, present when `return_complex=False`.